CBP Memo

```
library(tidyverse)
```

```
Warning: package 'dplyr' was built under R version 4.4.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.0     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(lubridate)
library(arrow)
```

```
Warning: package 'arrow' was built under R version 4.4.3
```

```

Attaching package: 'arrow'

The following object is masked from 'package:lubridate':

    duration

The following object is masked from 'package:utils':

    timestamp
```

```r
project_dir <- "/Users/simone/Desktop/DDP/cbp"

encounters_processed_dir <- file.path(
  project_dir, "data/encounters/processed"
)
apprehensions_processed_dir <- file.path(
  project_dir, "data/apprehensions/processed"
)
cross_reference_dir <- file.path(
  project_dir, "inputs/cross-reference-files"
)

apprehensions_latest_path <- file.path(
  apprehensions_processed_dir, "apprehensions-latest.parquet"
)
encounters_latest_path <- file.path(
  encounters_processed_dir, "encounters-latest.parquet"
)
cross_reference_file_path <- file.path(
  cross_reference_dir, "nationwide-encounters-fy20-fy23-aor.csv"
)

apprehensions_latest <- read_parquet(
  apprehensions_latest_path,
  col_select = c(apprehension_datetime, statute_charge))

encounters_latest <- read_parquet(
  encounters_latest_path,
  col_select = c(encounter_datetime, statute_charge))

cross_reference_file <- read_csv(cross_reference_file_path)
```

```
Rows: 53859 Columns: 12
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Month Grouping, Month (abbv), Component, Land Border Region, Area ...
dbl  (2): Fiscal Year, Encounter Count

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
encounters_edited <- encounters_latest |>
  mutate(
    title_of_authority = case_when(
      str_detect(statute_charge, "8 USC") ~ "Title 8",
      is.na(statute_charge) ~ "Title 42"
    )
  )

encounters_edited_v2 <- encounters_latest |>
  mutate(
    title_of_authority = case_when(
      !is.na(statute_charge) ~ "Title 8",
      is.na(statute_charge) ~ "Title 42"
    )
  )

apprehensions_edited <- apprehensions_latest |>
  mutate(
    title_of_authority = case_when(
      str_detect(statute_charge, "8 USC") ~ "Title 8",
      is.na(statute_charge) ~ "Title 42"
    )
  )
```

Intro

United States Border Patrol (USBP) releases apprehension and encounter data in separate sheets. While general encounters include all types of interactions at the border, Title 8 inadmissibles are recorded under the Office of Field Operations (OFO) rather than the USBP. Therefore, the USBP encounters datasets include only Title 8 apprehensions and Title 42 expulsions. On the other hand, USBP apprehensions data covers Title 8 apprehensions only.

Because intended difference between the USBP encounters datasets and the USBP apprehensions datasets is the pandemic-era Title 42 expulsions in the encounters dataset, we aim to consolidate encounters and apprehensions into one, comprehensive dataset that can be filtered depending on `title_of_authority`.

- Original logic:
  - If `statute_charge` contains 8 USC -> Title 8
  - If `statute_charge` is `NA` or missing -> Title 42
  - Otherwise -> NA

Unique Value Exploration

To consolidate these apprehensions and encounters datasets, we attempt to use the `statute_charge` field to indicate whether an encounter is a Title 8 apprehension or Title 42 expulsion. First, we explore the unique values in this field. *To be filled in after law student report?

Exploration

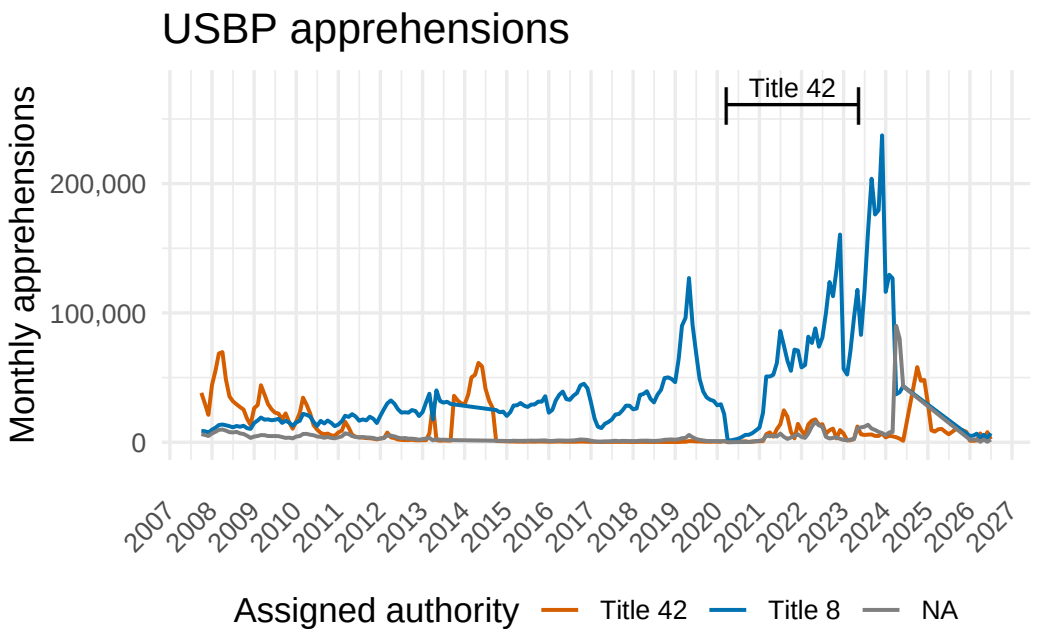
USBP Apprehension Counts Over Time

```
monthly_counts <- apprehensions_edited |>
  filter(!is.na(apprehension_datetime)) |>
  mutate(month = floor_date(as.Date(apprehension_datetime), "month")) |>
  count(month, title_of_authority)

bracket_y <- max(monthly_counts$n) * 1.1

ggplot(monthly_counts, aes(month, n, color = title_of_authority)) +
  geom_line(linewidth = 0.7) +
  annotate(
    "segment",
    x = as.Date("2020-03-20"), xend = as.Date("2023-05-11"),
    y = bracket_y, yend = bracket_y
  ) +
  annotate(
    "point",
    x = as.Date(c("2020-03-20", "2023-05-11")),
    y = bracket_y,
    shape = "|", size = 5
  ) +
  annotate(
    "text",
    x = as.Date("2021-10-14"), y = bracket_y * 1.05,
    label = "Title 42", size = 3.5
  ) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  scale_y_continuous(labels = scales::label_comma()) +
  scale_color_manual(
    values = c("Title 8" = "#0072B2", "Title 42" = "#D55E00")
  ) +
```

```
  labs(
    x = NULL, y = "Monthly apprehensions", color = "Assigned authority",
    title = "USBP apprehensions"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, vjust = 0.5))
```



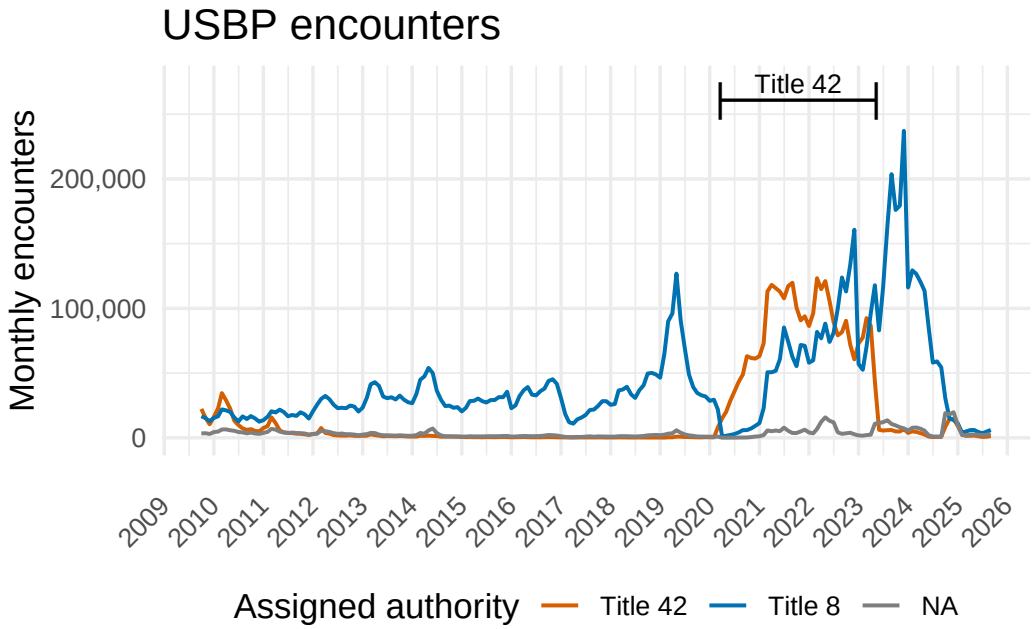
- I applied the same classification rule that labeled rows with 8 USC in `statute_charge` as Title 8, NA `statute_charge` as Title 42, and all else as NA to `apprehensions_latest.parquet`.
- These results lead me to think that `statute_charge` may not be a reliable indication of title of authority because that logic applied here implies that Title 42 exists in apprehensions, which we know not to be true.

USBP Encounter Apprehension Counts Over Time

```
monthly_counts <- encounters_edited |>
  filter(!is.na(encounter_datetime)) |>
  mutate(month = floor_date(as.Date(encounter_datetime), "month")) |>
  count(month, title_of_authority)

bracket_y <- max(monthly_counts$n) * 1.1
```

```r
ggplot(monthly_counts, aes(month, n, color = title_of_authority)) +
  geom_line(linewidth = 0.7) +
  annotate(
    "segment",
    x = as.Date("2020-03-20"), xend = as.Date("2023-05-11"),
    y = bracket_y, yend = bracket_y
  ) +
  annotate(
    "point",
    x = as.Date(c("2020-03-20", "2023-05-11")),
    y = bracket_y,
    shape = "|", size = 5
  ) +
  annotate(
    "text",
    x = as.Date("2021-10-14"), y = bracket_y * 1.05,
    label = "Title 42", size = 3.5
  ) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  scale_y_continuous(labels = scales::label_comma()) +
  scale_color_manual(
    values = c("Title 8" = "#0072B2", "Title 42" = "#D55E00")
  ) +
  labs(
    x = NULL, y = "Monthly encounters", color = "Assigned authority",
    title = "USBP encounters"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, vjust = 0.5))
```




```
# Encounter NA's based on new logic in Title 42 period
encounters_edited |> filter(between(
  as.Date(encounter_datetime),
  as.Date("2020-03-20"),
  as.Date("2023-05-11"))) |>
  summarise(sum(is.na(title_of_authority)))
```

```
# A tibble: 1 x 1
  `sum(is.na(title_of_authority))`
                             <int>
1                           155366
```

- The spike during the Title 42 period is promsising.
- There are 154680 NA values during the Title 42 period which could be Title 42 expulsions that we would never know about. How problematic would it be to not provide that info?

Title 8 Encounter and Apprehension Comparison

```
# Find overlapping date ranges
min(apprehensions_edited$apprehension_datetime)
```

```
[1] "2007-10-01 UTC"
```

```r
max(apprehensions_edited$apprehension_datetime)
```

```
[1] "2026-07-31 23:51:32 UTC"
```

```r
min(encounters_edited$encounter_datetime)
```

```
[1] "2009-10-01 00:01:00 UTC"
```

```r
max(encounters_edited$encounter_datetime)
```

```
[1] "2025-09-30 23:50:00 UTC"
```

```r
# Title 8 encounter count in comparable range
encounters_edited |> filter(between(
  as.Date(encounter_datetime),
  as.Date("2009-10-01"),
  as.Date("2025-09-30"))) |>
  summarise(sum(title_of_authority == "Title 8", na.rm = TRUE))
```

```
# A tibble: 1 x 1
  `sum(title_of_authority == "Title 8", na.rm = TRUE)`
                                                 <int>
1                                              8276838
```

```r
# Apprehension count in comparable range
apprehensions_edited |> filter(between(
  as.Date(apprehension_datetime),
  as.Date("2009-10-01"),
  as.Date("2025-09-30"))) |>
  summarize(n = n())
```

```
# A tibble: 1 x 1
        n
    <int>
1 9540874
```

```r
# Title 42 encounter count in Title 42 period
encounters_edited |> filter(between(
  as.Date(encounter_datetime),
  as.Date("2020-03-20"),
  as.Date("2023-05-11")))|>
  summarise(sum((title_of_authority == "Title 42"), na.rm = TRUE))
```

```
# A tibble: 1 x 1
  `sum((title_of_authority == "Title 42"), na.rm = TRUE)`
                                                    <int>
1                                                 3098730
```

```r
# Title 8 encounter count in Title 42 period
encounters_edited |> filter(between(
  as.Date(encounter_datetime),
  as.Date("2020-03-20"),
  as.Date("2023-05-11"))) |>
  summarise(sum(title_of_authority == "Title 8", na.rm = TRUE))
```

```
# A tibble: 1 x 1
  `sum(title_of_authority == "Title 8", na.rm = TRUE)`
                                                 <int>
1                                              2197266
```

```r
# Apprehension count in Title 42 period
apprehensions_edited |> filter(between(
  as.Date(apprehension_datetime),
  as.Date("2020-03-20"),
  as.Date("2023-05-11"))) |>
  summarize(n = n())
```

```
# A tibble: 1 x 1
        n
    <int>
1 2608509
```

- I counted apprehensions and Title 8 encounters within the comparable period (largest period of coverage shared by both datasets) and the Title 42 period.
- In the comparable period, the apprehensions dataset counts `9540874` apprehensions, while the encounters dataset counts `8276838` apprehensions. Assuming the apprehensions dataset is accurate, the logic under-counts apprehensions by `1264036`.

- In the Title 42 period, the apprehensions dataset counts 2608509 apprehensions, while the encounters dataset counts only 2197266 apprehensions. Assuming the apprehensions dataset is accurate during the Title 42 period, the logic under-counts apprehensions by 411243.

Title 42 CBP Dashboard Count Comparison

```
start_date <- as.Date("2020-03-20")
end_date <- as.Date("2023-05-11")

official_t42 <- cross_reference_file |>
  filter(
    Component == "U.S. Border Patrol",
    `Title of Authority` == "Title 42"
  ) |>
  mutate(
    month_number = match(
      str_to_upper(`Month (abbv)`),
      str_to_upper(month.abb)
    ),
    calendar_year = as.integer(`Fiscal Year`) -
      if_else(month_number >= 10L, 1L, 0L),
    month = make_date(calendar_year, month_number, 1)
  ) |>
  filter(between(
    month,
    floor_date(start_date, "month"),
    floor_date(end_date, "month")
  )) |>
  group_by(month) |>
  summarise(
    official_title42 = sum(`Encounter Count`, na.rm = TRUE),
    .groups = "drop"
  )

encounters_t42 <- encounters_edited |>
  filter(between(
    as.Date(encounter_datetime), start_date, end_date),
    title_of_authority == "Title 42"
  ) |>
  mutate(month = floor_date(as.Date(encounter_datetime), "month")) |>
```

```r
  count(month, name = "assigned_title42")

comparison <- official_t42 |>
  left_join(encounters_t42, by = "month") |>
  mutate(
    difference = assigned_title42 - official_title42,
    pct_difference = 100 * difference / na_if(official_title42, 0)
  )

print(comparison, n = Inf)
```

```
# A tibble: 39 x 5
   month      official_title42 assigned_title42 difference pct_difference
   <date>                <dbl>            <int>      <dbl>          <dbl>
 1 2020-03-01             7094             7181         87          1.23
 2 2020-04-01            15018            14966        -52         -0.346
 3 2020-05-01            20084            20067        -17         -0.0846
 4 2020-06-01            28534            28731        197          0.690
 5 2020-07-01            35444            35782        338          0.954
 6 2020-08-01            42808            43193        385          0.899
 7 2020-09-01            48389            48874        485          1.00
 8 2020-10-01            63018            63049         31          0.0492
 9 2020-11-01            61354            61635        281          0.458
10 2020-12-01            60620            61168        548          0.904
11 2021-01-01            62545            63052        507          0.811
12 2021-02-01            72335            72944        609          0.842
13 2021-03-01           107278           113082       5804          5.41
14 2021-04-01           110869           118157       7288          6.57
15 2021-05-01           111542           115622       4080          3.66
16 2021-06-01           103096           112937       9841          9.55
17 2021-07-01            94264           107733      13469         14.3
18 2021-08-01            93158           117230      24072         25.8
19 2021-09-01           100616           119793      19177         19.1
20 2021-10-01            92741           100604       7863          8.48
21 2021-11-01            87913            90735       2822          3.21
22 2021-12-01            79873            93880      14007         17.5
23 2022-01-01            77541            86354       8813         11.4
24 2022-02-01            90934            96277       5343          5.88
25 2022-03-01           109702           123271      13569         12.4
26 2022-04-01            98362           114836      16474         16.7
27 2022-05-01           103624           121077      17453         16.8
28 2022-06-01            92814           105820      13006         14.0
```

29	2022-07-01	75800	89485	13685	18.1
30	2022-08-01	72711	79111	6400	8.80
31	2022-09-01	72494	81621	9127	12.6
32	2022-10-01	80097	90414	10317	12.9
33	2022-11-01	69101	71816	2715	3.93
34	2022-12-01	51385	60600	9215	17.9
35	2023-01-01	66538	72951	6413	9.64
36	2023-02-01	76209	77376	1167	1.53
37	2023-03-01	91160	92468	1308	1.43
38	2023-04-01	84203	86496	2293	2.72
39	2023-05-01	31585	38342	6757	21.4

- Percent differences seem negligible at first but generally grow by month throughout the Title 42 period.

Title 42 CBP Dashboard Count Comparison V2

```
start_date <- as.Date("2020-03-20")
end_date <- as.Date("2023-05-11")

official_t42 <- cross_reference_file |>
  filter(
    Component == "U.S. Border Patrol",
    `Title of Authority` == "Title 42"
  ) |>
  mutate(
    month_number = match(
      str_to_upper(`Month (abbv)`),
      str_to_upper(month.abb)
    ),
    calendar_year = as.integer(`Fiscal Year`) -
      if_else(month_number >= 10L, 1L, 0L),
    month = make_date(calendar_year, month_number, 1)
  ) |>
  filter(between(
    month,
    floor_date(start_date, "month"),
    floor_date(end_date, "month")
  )) |>
  group_by(month) |>
  summarise(
```

```
    official_title42 = sum(`Encounter Count`, na.rm = TRUE),
    .groups = "drop"
  )

encounters_t42 <- encounters_edited_v2 |>
  filter(between(
    as.Date(encounter_datetime), start_date, end_date),
    title_of_authority == "Title 42"
  ) |>
  mutate(month = floor_date(as.Date(encounter_datetime), "month")) |>
  count(month, name = "assigned_title42")

comparison <- official_t42 |>
  left_join(encounters_t42, by = "month") |>
  mutate(
    difference = assigned_title42 - official_title42,
    pct_difference = 100 * difference / na_if(official_title42, 0)
  )

print(comparison, n = Inf)
```

```
# A tibble: 39 x 5
   month      official_title42 assigned_title42 difference pct_difference
   <date>                <dbl>            <int>      <dbl>          <dbl>
 1 2020-03-01             7094             7181         87          1.23
 2 2020-04-01            15018            14966        -52         -0.346
 3 2020-05-01            20084            20067        -17         -0.0846
 4 2020-06-01            28534            28731        197          0.690
 5 2020-07-01            35444            35782        338          0.954
 6 2020-08-01            42808            43193        385          0.899
 7 2020-09-01            48389            48874        485          1.00
 8 2020-10-01            63018            63049         31          0.0492
 9 2020-11-01            61354            61635        281          0.458
10 2020-12-01            60620            61168        548          0.904
11 2021-01-01            62545            63052        507          0.811
12 2021-02-01            72335            72944        609          0.842
13 2021-03-01           107278           113082       5804          5.41
14 2021-04-01           110869           118157       7288          6.57
15 2021-05-01           111542           115622       4080          3.66
16 2021-06-01           103096           112937       9841          9.55
17 2021-07-01            94264           107733      13469         14.3
18 2021-08-01            93158           117230      24072         25.8
```

19	2021-09-01	100616	119793	19177	19.1
20	2021-10-01	92741	100604	7863	8.48
21	2021-11-01	87913	90735	2822	3.21
22	2021-12-01	79873	93880	14007	17.5
23	2022-01-01	77541	86354	8813	11.4
24	2022-02-01	90934	96277	5343	5.88
25	2022-03-01	109702	123271	13569	12.4
26	2022-04-01	98362	114836	16474	16.7
27	2022-05-01	103624	121077	17453	16.8
28	2022-06-01	92814	105820	13006	14.0
29	2022-07-01	75800	89485	13685	18.1
30	2022-08-01	72711	79111	6400	8.80
31	2022-09-01	72494	81621	9127	12.6
32	2022-10-01	80097	90414	10317	12.9
33	2022-11-01	69101	71816	2715	3.93
34	2022-12-01	51385	60600	9215	17.9
35	2023-01-01	66538	72951	6413	9.64
36	2023-02-01	76209	77376	1167	1.53
37	2023-03-01	91160	92468	1308	1.43
38	2023-04-01	84203	86496	2293	2.72
39	2023-05-01	31585	38342	6757	21.4